Mechanism Design for Assisting Shoulder Abduction using Origami Petal Fold driven by Shape Memory Alloy (SMA)-Based Artificial Muscle

Chongyoung Chung, Kyujin Hyeon, Jaeyeon Jeong, and Ki-Uk Kyung*, *Member, IEEE*

***Abstract*— This paper proposes a novel mechanism to assist shoulder abduction using a petal mechanism based on petal fold origami driven by shape memory alloy (SMA)-based artificial muscle. To follow the scapulohumeral rhythm, the synchronized movement of the scapula and humerus during shoulder abduction, the proposed mechanism uses two petal structures to mimic the scapula and humerus, respectively, and it is called the double-petal mechanism. This mechanism is designed to not only make large torque output using high mechanical advantage but also reduce the compressive force to the glenohumeral joint. With a petal mechanism, an artificial muscle using bundles of SMA spring actuators, a lightweight powerful actuator, is used to reduce the overall weight and make large output force. While the SMA spring actuator has restoring force without energy input, the proposed robot can assist both passively and actively according to the required torque. The proposed robot can generate up to 6.36 Nm passively without energy input, only using restoring force of the actuator, and 12.6 Nm actively with energy input at a 90° abduction angle.**

I. Introduction

Wearable robots have been widely developed in various fields such as the manufacturing industry, assisting in the activities of daily living (ADL), and rehabilitation or treatment [1]-[3]. One of the most interesting movements of the wearable robot is shoulder abduction because of its importance and high-frequency usage. While wearable robots for shoulder abduction move with the upper limb of the users, there are several requirements for safe human-robot interaction and sufficient assistance. First, the maximum 10 Nm torque output to lift the overall upper limb weight are necessary [2]. And, to avoid rotator cuff tear problems due to the friction between the rotator cuff and acromion, the compressive force to the glenohumeral (GH) joint during assistance needs to be minimized. Also, when a wearable robot assists the shoulder abduction, it does not simply rotate at a point like a ball-socket joint, instead, it needs to consider the scapulohumeral rhythm (SHR), a special rhythm that rotates together with a ratio of 1:2 between scapula and humerus, to avoid misalignment problem [3]. However, many research is mainly focused on large torque output rather than safety interactions such as compressive force and following SHR.

A petal mechanism has a high potential as a candidate for a mechanism for assisting shoulder abduction. The petal mechanism is inspired by petal fold origami as shown in Fig. 1 (a) and (b) [4]. The working principle of this mechanism is the same as petal fold origami; the actuator is placed and contracted between E and F ($\overline{EF}$ ↓), which are both ends of the wing, and the structure is straightened and unfolded ($\angle AOC$ ↑). One of the main advantages is a small compressive force because the rotation axis and the direction of the actuator force ($\overline{EF}$) are parallel. Another advantage is a high mechanical advantage (MA) due to the scissors mechanism structure (□AECF) inside the petal mechanism.

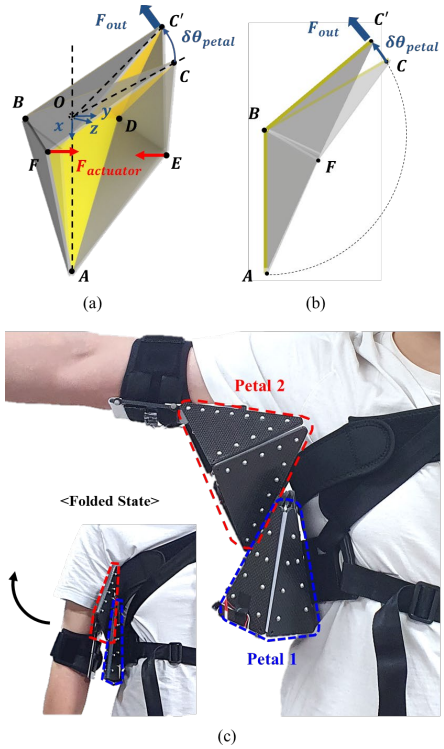


Fig. 1. (a) Diagonal and (b) side view of the working principle of a single petal mechanism. When actuator contracted, the mechanism is straightening the bending and unfolding. (c) Conceptual robot uses a double-petal mechanism which is composed of petal 1 and 2, and it is in the armpit and assist the shoulder abduction while the actuator contracted.

Shape memory alloy (SMA) actuators are good as a contraction and linear actuator to work with the petal

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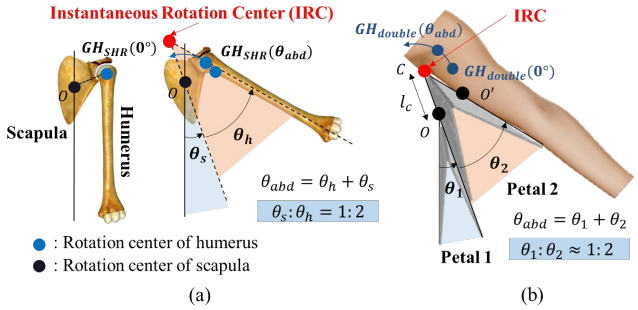


Fig. 2. Instantaneous rotation center (IRC) of (a) the following SHR and (b) double-petal mechanism. The double-petal mechanism mimics the scapula and humerus, and especially the IRC.

mechanism. This actuator shows good performances such as high force output due to the high-power density [5], [6]. However, since the wire geometry-based SMA actuator shows a high power-to-weight ratio (> 50 W/g), the strain is limited in low levels (< 5 %) [7]. Therefore, to improve the strain (> 200 %), the spring geometry-based SMA actuator was selected for the proposed mechanism. Also, since the SMA spring actuator can generate a restoring force without energy input, the energy-efficient assistance of shoulder abduction is possible even without a current input. And this actuator geometry itself contracts, so it is beneficial to contract the petal mechanism directly using this actuator.

In this paper, we propose a new mechanism using an origami-based petal mechanism, to assist shoulder abduction driven by SMA. The proposed petal mechanism is a new mechanism that mimics the SHR composed of the scapula and humerus, to support the shoulder abduction. Therefore, the double-petal mechanism using two petal mechanisms is used to mimic the scapula and humerus. An optimization process to follow the SHR is conducted to decide the geometry of the petal mechanism and specifications of the SMA spring actuator. After design the mechanism, the wearable robot using this mechanism and actuator was developed and it assists the shoulder abduction as shown in Fig. 1 (c).

II. Design Process of the Mechanism for Shoulder Abduction

A. Mathematical Modeling of the Mechanism

Before the mechanism design, the scapulohumeral rhythm (SHR) during shoulder abduction, was examined as shown in Fig. 2 (a). The SHR can be simply modeled as a rotation of two triangles with a variable dislocation length as the abduction angle and an instantaneous rotation center (IRC) of SHR is the intersection point of each line. Therefore, the proposed mechanism is modeled as a double-petal mechanism composed of two petal mechanisms with a dislocation length (l_c) to follow the SHR as shown in Fig. 2 (b). To minimize the misalignment error of the GH joint between SHR and the double-petal mechanism, the objective function is defined as (1). And the objective function was simulated to find optimized l_c using each GH joint trajectory model as (2) and (3) and 70 mm was selected as l_c. The $R(\theta)$ in (1)-(3) means that the rotation matrix rotates about θ in a plane.

$$err(\theta_{abd}) = \|GH_{SHR}(\theta_{abd}) - GH_{double}(\theta_{abd})\| \quad (1)$$

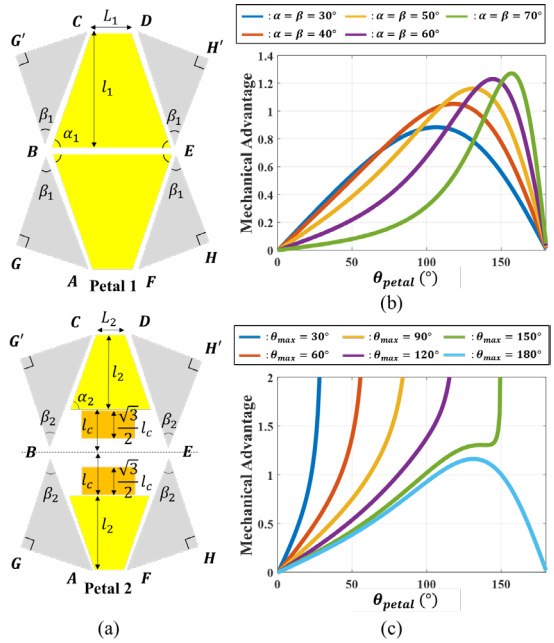


Fig. 3. (a) Development figures of petal 1 and petal 2. (b) The mechanical advantage relationships about (b) $\boldsymbol{\alpha}(=\boldsymbol{\beta})$ (c) and $\boldsymbol{\theta_{max}}$ when $\boldsymbol{\alpha} = \mathbf{50°}$.

$$\overrightarrow{GH}_{SHR}(\theta_{abd}) = R(\theta_s)\cdot\overrightarrow{GH}_{SHR}(0°) = R(\frac{\theta_{abd}}{3})\cdot\overrightarrow{GH}_{SHR}(0°) \quad (2)$$

$$\begin{aligned}\overrightarrow{GH}_{double}(\theta_{abd}) &= \overrightarrow{OC}(\theta_1) + \overrightarrow{C,GH}_{double}(\theta_{abd}) \\ &= R\left(\frac{\theta_{abd}}{3}\right)\cdot\overrightarrow{OC}(0°) + R(\theta_{abd})\cdot\overrightarrow{C,GH}_{double}(0°)\end{aligned} \quad (3)$$

Then, the geometries of the double-petal mechanism are optimized to maximize the MA and maintain the angle ratio ($=\theta_2/\theta_1$) of 2, where θ_1 and θ_2 are the rotation angles of petal 1 and 2, respectively. The development figures of each petal mechanism are shown in Fig. 3 (a). The geometries of the two petals are different and the center part of petal 2 is removed, instead, the small rectangles are positioned. This is because the rotation part of $\triangle OO'C$ as shown in Fig. 2 (b) can collide with a body, the $\triangle OO'C$ part was inside the reverse fold using two rectangles. The value of α_i and β_i determine the maximum rotation angle ($\theta_{max,i}$). The relationship between α_i and β_i can be calculated as (4) using the parallel property of $\overrightarrow{AG}//\overrightarrow{CG}$ when petal rotates θ_{max} as shown in Fig. 3 (a) in 3D space. Therefore, β_i can be expressed as a function of α_i and θ_{max}.

$$\beta_i = sin^{-1}\left(sin\alpha_i \cdot sin\frac{\theta_{max,i}}{2}\right) = f(\alpha_i, \theta_{max,i}) \quad (4)$$

Since the petal 1 and 2 are needed to limit the rotation only up to $60°$ ($0° \le \theta_1 \le 60°$) and $120°$ ($0° \le \theta_2 \le 120°$), respectively, β_1 and β_2 can be calculated as (5) and (6).

$$\beta_1 = sin^{-1}\left(sin\alpha_1 \cdot sin\frac{\theta_{max}}{2}\right) = sin^{-1}\left(\frac{sin\alpha_1}{2}\right) \quad (5)$$

$$\beta_2 = sin^{-1}\left(sin\alpha_2 \cdot sin\frac{\theta_{max}}{2}\right) = sin^{-1}\left(\frac{\sqrt{3}}{2}sin\alpha_2\right) \quad (6)$$

Also, the value of α_i and β_i determine the MA of petal mechanism as (8) using the orthogonality of $\overrightarrow{AG}\cdot\overrightarrow{CG} = 0$ as (7). Therefore, the mechanical advantage can be finally expressed as a function of α_i and θ_{petal}.

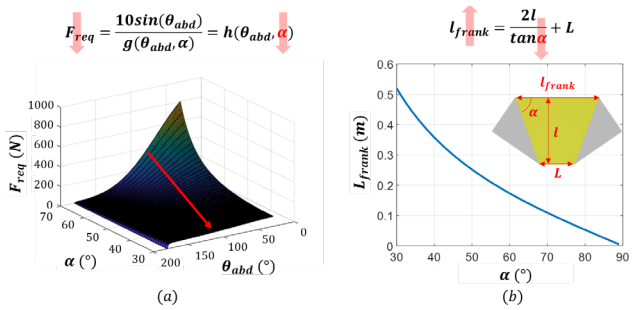


Fig. 4. The trade-off relationships of the (a) required force to each petal mechanism and (b) the overall frank length. Both required force and frank length should be minimized.

$$cos(\theta_{\angle FBD}) + tan(\alpha_i) \cdot cos(\frac{\theta_{petal}}{2}) \cdot sin(\theta_{\angle FBD}) = \frac{cos(\beta_i)}{cos(\alpha_i)} \quad (7)$$

$$MA = \frac{F_{out}}{F_{in}} = -\frac{\delta\overline{EF}}{\delta\widehat{AOC}} = -\frac{2cos(\beta_i)}{sin(\alpha_i)} \cdot \frac{dcos(\theta_{\angle FBD})}{d\theta_{petal}} = f(\alpha_i, \theta_{petal}) \quad (8)$$

Fig. 3 (b) and (c) show the MA properties about α_i, β_i, and θ_{petal}. If $\alpha_i = \beta_i$, which means that $\theta_{max,i} = 180°$, the MA is increased as the α_i increases. If $\theta_{max,i}$ is changed to a value lower than 180° such as 120°, the MA trends change at the same α as shown in Fig. 3 (c). The double-petal mechanism should maintain the angle ratio of 2 to mimic the SHR, which means that the torque output of each petal should same when the angle ratio is maintained as 2. If α_1 and α_2 are the same as α, the force output ($F_{out,i}$) are the same when the same force input ($F_{actuator,i}$) is exerted on each petal while the $\theta_1 : \theta_2 = 1:2$ as shown in Fig. 3 (c). Also, if l_1 and $l_2 + l_c$ are the same as l, the torque output will be also the same because the moment arm is the same as (9)-(11).

$$T_1 = l_1 \cdot F_{out,1} = l_1 \cdot F_{actator,1} \cdot MA(\alpha_1, \theta_1) \quad (9)$$

$$T_2 = (l_2 + l_c) \cdot F_{out,2} = (l_2 + l_c) \cdot F_{actuator,2} \cdot MA(\alpha_2, \theta_2) \quad (10)$$

$$T_1(l, \alpha, \theta_1) = T_2(l, \alpha, \theta_2) \quad where\ \theta_2 = 2\theta_1 \quad (11)$$

B. Mechanism Optimization

In the case of l, as a moment arm, the larger it is, the better it is in terms of torque output. However, if it is longer than the length of the upper arm, it may interfere with the movement of other joints, so it is assumed as 150 mm to match the length of the upper arm. Next, α is a variable that determines the MA as (9)-(11). To determine this, the required force (F_{req}) to be applied through the actuator was obtained, and the force to be applied through the actuator decreases as α decreases as shown in Fig. 4 (a). However, if the α becomes smaller, the length of the petal located in the armpit (L_{frank}) increases as shown in Fig. 4 (b). An increase in this length means an increase in the overall structure size, thus creating a trade-off relationship with the required force. To optimize the two values, we multiplied each other to create an objective function and simulated to minimize it: $f_{obj} = F_{req} \times L_{frank}$. The results show that the objective function is optimized at $\alpha = 52°$. Then, the β_1 and β_2 can be calculated using equation (5) and (6) as 23° and 43°, respectively. Based on this result, it was possible to derive the force that should be generated through the shape memory alloy, and it was found that a force of up to 200 N should be generated.

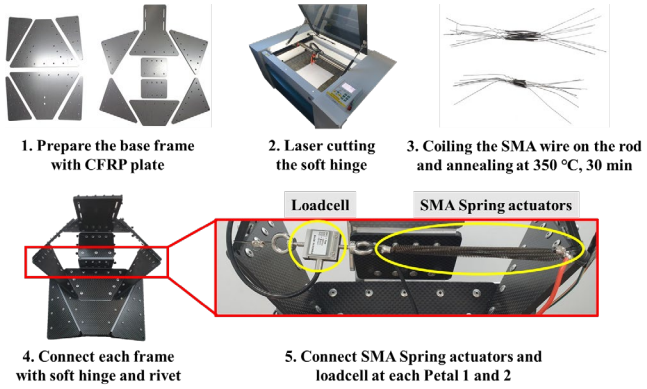


Fig. 5. Fabrication process of the proposed mechanism.

TABLE I. Design Parameters of the SMA spring actuator

Parameter	Value	
	Petal 1	Petal 2
Wire diameter (d)	0.5 mm	0.5 mm
Coil outer diameter (D)	2.0 mm	2.0 mm
Number of turns (n)	40	100
Number of coils (N)	10	10

Then, the artificial muscles using SMA spring actuators were designed. To maximize the restoring force, the lower transition temperature from the martensite to austenite (A_s) than the ambient temperature was selected about 26.4 ℃. And the bundles of SMA spring actuators were used to make more than the maximum 200 N force output. Then, the overall design parameters of the SMA spring actuator can be determined as shown in TABLE I based on our previous research [3], [5], [6].

C. Fabrication Process of the Mechanism

Fig. 5 shows a fabrication process of the proposed mechanism. First, the base frame is fabricated using 2 mm thickness carbon fiber-reinforced plastic (CFRP) by a CNC machine following the optimized design in the previous section. Then, a soft hinge composed of a composite layer with PET film (0.1mm) and nylon cloth is fabricated by a laser cutter. Then, the SMA wire is coiled on the rod and annealed for 30 mins at 350℃ in the oven and quenched in the cold water. After preparing the materials, the frames are connected with the soft hinge using rivets. Finally, the SMA spring actuators and loadcell (Dacell, UU74) are connected at the end of each petal structure, respectively. The loadcell will be used to control by measuring the force of the actuator.

III. Performance Evaluation

A. Experimental Setup

After optimization and fabrication of the proposed mechanism, the performance evaluation is needed to validate the theoretical data. Therefore, the two experiments are conducted in static and dynamic conditions, respectively. There are two things that are needed to validate in static condition; a comparison of the torque output between theoretical and experimental data and geometric constraints that the angle ratio is maintained as 2 when the same forces are exerted at each petal mechanism. To validate this, the angle

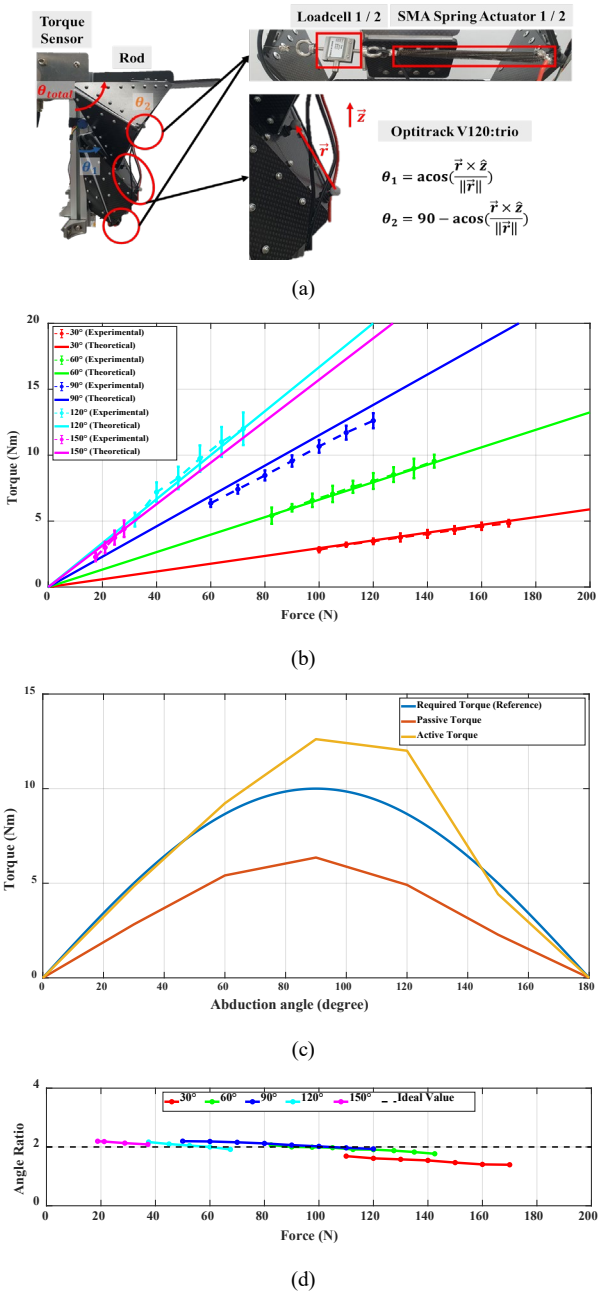


Fig. 6. (a) The test bed for measuring torque output and angle ratio as the input forces to each petal increases. (b) Comparison of the input force and torque output relationship between theoretical data (solid line) and experimental data (dotted line). (c) Relationship between the torque output and abduction angle. (d) Relationship between the angle ratio and input force as the total angle.

ratio and torque output were measured while increasing the force equally by applying current to each actuator.

Fig. 6 (a) shows the experimental setup for performance evaluation in static condition. To fix the total angle in a static condition at specific angles (30°, 60°, 90°, 120°, and 150°), the rod is used. After fixing the total angle (θ_{total}), the same forces are exerted to each SMA spring actuators by measuring the forces of the actuator using loadcell. Then, each petal angle is measured using motion capture camera (Optitrack, V120:trio and the torque sensor (Dacell, TCN16) is measured of the torque output due to the force of the actuator.

B. Performance of the Mechanism in Static Condition

First, the experimental torque output data is compared to theoretical data as shown in Fig. 6 (b). The input force and torque output at each angle are plotted in x-axis and y-axis, respectively. The result shows that the error between the experimental and theoretical data of 30° and 60° are below the 10 %. Also, the maximum torque at 90° is about 12.6 Nm and 6.36 Nm, which are actively and passively, respectively.

Next, Fig. 6 (d) shows the trend of the angle ratio as the input forces (F_{in}) of the SMA actuators increase at each total angle (θ_{total}). The angle ratio of the mechanism is maintained at 2 which is an error of 10 % at the most total angle. However, in the case of $\theta_{total} = 30°$, the angle ratio maintains about 1.5. Therefore, the established geometric constraint that the angle ratio will maintain a value of 2 was validated in most total angles except for the sections below $\theta_{total} = 30°$.

IV. Conclusion & Future Works

We proposed and designed a new wearable mechanism to assist with shoulder abduction movement based on the petal mechanism. To mimic the scapula and humerus of the SHR, the double-petal mechanism is used to follow each bone. Based on this mechanism, assisting shoulder abduction maintaining the angle ratio of 2 following SHR can be mimicked. Also, the proposed robot can assist the shoulder abduction up to 12.6 Nm and 6.36 Nm with and without the energy input, respectively. However, there are several issues to resolve in the future. First, the proposed robot was designed to assist only with shoulder abduction. It means that the proposed mechanism can hinder shoulder flexion/extension because it limits the degree of freedom (DOF). Therefore, in the future, we plan to add a simple origami-based mechanism so that it does not disturb the corresponding other DOF even without placing additional actuators. Also, after developing the wearable robot, the user test to verify the assistive effect of the proposed wearable robot is required. To validate the assistive effect, the surface electromyography (sEMG) of the lateral deltoid will be measured with and without the assistance of the wearable robot.

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